

# The Discriminant Resolvent of the $A_5$ -Cubic Pencil

Tavis Rudd

August 2026

## Abstract

The nonstandard  $A_5$ -invariant pencil of cubic threefolds carries five principal gluing kernels: three rational and two golden. We prove that this is not an accidental  $3 + 2$  count. The five kernels are the two proper transitive resolvents of the  $S_3$ -torsor of ordered nonzero two-torsion on the relative elliptic norm axis. The rational triple is the root cover and the golden pair has the sign character, hence is the discriminant double cover of the same two-division cubic.

Writing  $t$  for the signed cubic parameter and  $T = 81t^2$ , the unmarked elliptic period is the standard  $X_0(3)$ -coordinate

$$j = \frac{(T + 27)(T + 3)^3}{T}.$$

The signed cubic line is the sign modular curve  $r^2 = T$ , with  $r = 9t$  after choosing its deck convention. The degree-three resolvent is  $X_0(6)$ , and their common degree-six Galois closure belongs to  $\Gamma_0(3) \cap \Gamma(2)$ . We give rational equations for all three covers and distinguish the two modular cusps from the additional interior boundary points where the cubic degenerates. In particular the modular identification concerns the actual relative principal-kernel packet, not only an auxiliary elliptic curve with the same  $j$ -invariant. Over the signed line, the resulting cyclic cubic cover is totally ramified exactly at the two fibres with five  $A_2$ -points. At the chordal value  $h = 5$ , the hyperelliptic limit has Jacobian isogenous to the fifth power of the elliptic curve selected by the same norm-axis coordinate.

**Keywords.** cubic threefold; intermediate Jacobian; elliptic two-torsion; discriminant resolvent; modular curve;  $A_5$ .

## 1 The result

Let  $W \subset \mathbf{C}^6$  be the hyperplane  $\sum_{i=0}^5 z_i = 0$ . The nonstandard transitive copy of  $A_5 \subset S_6$  has two orbits  $\mathcal{T}_+$  and  $\mathcal{T}_-$ , each of size ten, on the three-element subsets of  $\{0, \dots, 5\}$ . Put

$$P = \sum_{i=0}^5 z_i^3, \quad Q = \sum_{I \in \mathcal{T}_+} \prod_{i \in I} z_i - \sum_{I \in \mathcal{T}_-} \prod_{i \in I} z_i, \quad X_t = \{P + tQ = 0\} \subset \mathbf{P}(W).$$

The outer coset of the normalizer  $S_5 = N_{S_6}(A_5)$  fixes  $P$ , negates  $Q$ , and therefore sends  $X_t$  to  $X_{-t}$ . Thus  $t$  is the signed parameter and  $t^2$  is the unmarked parameter.

We use the following precise input from the six-axis construction.

**Proposition 1.1** (Programme input). *Over the connected smooth marked cubic base  $B^\circ$ , there are a relative intermediate Jacobian  $\mathcal{J}$ , an elliptic norm axis  $\mathcal{E}$ , and an  $A_5$ -equivariant isogeny*

$$f : \mathcal{E} \otimes (\mathbf{Z}^\Omega / \mathbf{Z}\mathbf{1}) \longrightarrow \mathcal{J}$$

whose pulled-back polarization is the form induced by  $6I_6 - J_6$ . The  $A_5$ -stable maximal isotropic halves of its two-primary discriminant form the five-sheet local system

$$\mathcal{P} = \mathbf{P}_{\mathbf{F}_4}(\mathbf{F}_4 \otimes_{\mathbf{F}_2} \mathcal{E}[2]). \quad (1.1)$$

The actual subgroup  $\ker(f)_{(2)}$  is one section of this candidate packet, and it selects one of the two non- $\mathbf{F}_2$ -rational sheets.

*Proof.* This is exactly [8, Relative six-axis source and Principal gluing packet]. The first result constructs  $f$ , proves the polarization identity, and identifies the two-primary discriminant local system. The second classifies its five stable halves and states separately which half is selected by the actual relative kernel.  $\square$

Proposition 3.1 below identifies the elliptic curve used in the period calculation with this norm axis, rather than only with an elliptic factor in its isogeny class.

**Theorem 1.2** (Cubic-modular resolvent). *Set  $T = 81t^2$ . Then the following assertions hold over the smooth cubic base.*

1. *The elliptic norm axis has*

$$j(\mathcal{E}) = J(T) := \frac{(T + 27)(T + 3)^3}{T}.$$

*Consequently the unmarked period map identifies its normalized coarse base with an open subcurve of  $X_0(3)$ .*

2. *The packet  $\mathcal{P}$  is the disjoint union of the root cover  $\mathbf{P}(\mathcal{E}[2])$ , of degree three, and a degree-two cover having the sign character of  $\text{Gal}(\mathcal{E}[2]) \subset S_3$ . The second cover is the discriminant resolvent of the two-division cubic.*
3. *Over  $X_0(3)$ , the root cover is  $X_0(6)$ , the sign cover has function field  $\mathbf{Q}(T)(\sqrt{T})$ , and their common Galois closure belongs to  $\Gamma_0(3) \cap \Gamma(2)$ . After one choice of deck involution, the signed cubic coordinate identifies with the sign coordinate by*

$$r^2 = T, \quad r = 9t.$$

4. *The compact sign curve has genus zero, two cusps of widths 2 and 6, and two elliptic points of order three. Its map to  $X_0(3)$  is ramified at the two cusps and nowhere else as a coarse map.*
5. *The geometric mod-two monodromy of the actual norm axis over the marked smooth cubic base is exactly  $A_3 \simeq C_3$ .*
6. *Over the signed cubic line, the cyclic cubic splitting cover is totally ramified exactly at the two points  $t^2 = -1/3$  and is unramified at every other cubic boundary point.*

*The smooth cubic base is obtained from the sign modular curve by deleting, in addition to the cusps over  $T = 0, \infty$ , the points over  $T = -27$  and  $T = 729/5$ . Those last two values are interior modular points, not cusps.*

The theorem identifies local systems and their monodromy. It does not say that a fivefold gluing is literally an elliptic 2-isogeny, nor that the compactified cubic pencil is the modular stack.

## 2 The finite $3 + 2$ resolvent

Let  $V$  be a two-dimensional  $\mathbf{F}_2$ -vector space and  $V_4 = \mathbf{F}_4 \otimes_{\mathbf{F}_2} V$ . Scalar extension gives a distinguished three-element subset

$$\mathbf{P}_{\mathbf{F}_2}(V) \subset \mathbf{P}_{\mathbf{F}_4}(V_4).$$

Write  $D(V)$  for its two-element complement.

**Lemma 2.1** (Sign lemma). *Under  $\mathrm{GL}(V) \simeq S_3$ , the action on  $D(V)$  is the sign of the natural action on  $\mathbf{P}(V)$ . Thus the two finite  $S_3$ -sets are*

$$\mathbf{P}(V) \simeq S_3/C_2, \quad D(V) \simeq S_3/A_3.$$

*Proof.* Choose a basis and write

$$\mathbf{P}^1(\mathbf{F}_4) = \{\infty, 0, 1, \omega, \omega^2\}, \quad \mathbf{P}^1(\mathbf{F}_2) = \{\infty, 0, 1\}.$$

The transformations  $x \mapsto x + 1$  and  $x \mapsto 1/(x + 1)$  generate  $\mathrm{PGL}_2(\mathbf{F}_2)$ . The first acts as  $(0 \ 1)(\omega \ \omega^2)$ ; the second cycles  $(\infty \ 0 \ 1)$  and fixes  $\omega, \omega^2$ . The first generator is odd on each of the three-set and the two-set, while the second is even on both. This proves the assertion.  $\square$

For a degree-three finite *étale* cover  $C \rightarrow S$ , let  $\mathrm{Or}(C/S)$  denote the double cover associated to the sign of its monodromy in  $S_3$ . Lemma 2.1 proves

$$[D(V)] = [\mathrm{Or}(\mathbf{P}(V)/S)] \quad \text{in } H^1(S, C_2). \quad (2.1)$$

This equality of torsor classes is canonical. A sheetwise isomorphism of the two covers has two choices, interchanged by the deck involution. This minor distinction is what makes the normalization  $r = 9t$ , rather than  $r = -9t$ , a convention.

There is a useful higher-prime form of this finite geometry. If  $V$  has dimension two over  $\mathbf{F}_\ell$ , then, as  $\mathrm{PGL}(V)$ -sets,

$$\mathbf{P}^1(\mathbf{F}_{\ell^2}) = \mathbf{P}^1(\mathbf{F}_\ell) \sqcup (\mathbf{P}^1(\mathbf{F}_{\ell^2}) \setminus \mathbf{P}^1(\mathbf{F}_\ell)) \simeq \mathrm{PGL}(V)/B \sqcup \mathrm{PGL}(V)/C_{\mathrm{ns}},$$

where  $B$  is a Borel subgroup and  $C_{\mathrm{ns}}$  a nonsplit Cartan. Frobenius pairs the second orbit, and its quotient is  $\mathrm{PGL}(V)/N_{\mathrm{ns}}$ , where  $N_{\mathrm{ns}}$  is the Cartan normalizer; compare the modular interpretation in [2, Section 2]. For  $\ell = 2$ , the normalizer is all of  $\mathrm{PGL}_2(\mathbf{F}_2)$ , while  $C_{\mathrm{ns}} = A_3$ . Thus only at two does the entire nonsplit-Cartan orbit reduce to a double cover, namely the sign torsor above.

Now let an elliptic curve be given locally, with 2 invertible, by

$$E : y^2 = x^3 + Ax + B.$$

The nonzero two-torsion points are the roots of  $f_2(x) = x^3 + Ax + B$ . The orientation cover of this root cover is therefore

$$u^2 = \mathrm{disc}(f_2), \quad \mathrm{disc}(f_2) = -4A^3 - 27B^2.$$

Since  $\Delta_E = 16 \mathrm{disc}(f_2)$ , it has the same square class as the elliptic discriminant. Invariantly,  $\Delta_E$  is a nowhere-vanishing section of the twelfth power of the Hodge line; the double cover is the torsor of its square roots. No global scalar equation for  $\Delta_E$  is implied before a Hodge trivialization is chosen.

Applying the lemma to (1.1) proves Theorem 1.2(2). It also gives a useful monodromy consequence: choosing one golden sheet reduces the mod-two image from  $S_3$  to  $A_3 \simeq C_3$ .

### 3 The cubic parameter and the actual norm axis

The period calculation starts with the Prym quotient used by van Geemen and Yamauchi. In their coordinate  $u$ , two auxiliary parameters in the Weierstrass model of its elliptic factor are

$$a(u) = -\frac{32(u-3)^4}{9(u+1)^3(u+3)^2}, \quad b(u) = -\frac{8(u-3)^2(u-1)}{(u+1)(u+3)^2}.$$

The proof of [9, Proposition 3.1] gives the generalized Weierstrass coefficients

$$\begin{aligned} a_1 &= b^2/4 + b - 4, & a_3 &= (a+b)^2/2 + 4a + 6b - 8, \\ a_2 &= 2 - a - ab/2 - b^4/64 - b^3/8 - b^2/4 - b, & a_4 &= 4(a+b-1), \\ a_6 &= (a+b-1)(8-4a-2ab-b^4/16-b^3/2-b^2-4b). \end{aligned}$$

Here is the coordinate comparison. Index five of the six coordinates by  $k \in \mathbf{F}_5$ , index the last by  $\infty$ , choose a fifth root  $\zeta$ , and put

$$z_k = \frac{1}{5} \left( X_0 + \sum_{m=1}^4 \zeta^{mk} X_m \right), \quad z_\infty = -X_0.$$

If

$$c = 2\zeta^3 + 2\zeta^2 + 1,$$

then  $c^2 = 5$ . With  $s = ct$ , multiplication of  $P+tQ$  by  $25/3$  and the reordering  $(x, u, v, z, y) = (X_4, X_3, X_2, X_1, X_0)$  give

$$\begin{aligned} &(1-s/3)(xu^2 + zv^2) + 2(1+s)yuv \\ &+ (1+s/3)(z^2u + x^2v) - 8y^3 + 2(1-s)xyz. \end{aligned}$$

Set  $A_- = 1 - s/3$ ,  $A_+ = 1 + s/3$ . Replacing  $x, z, y$  by

$$\frac{2A_-}{A_+}x, \quad \frac{2A_-}{A_+}z, \quad \frac{2A_-^2}{A_+(1+s)}y$$

and multiplying the equation by  $A_+/(2A_-^2)$  puts it in the van Geemen–Yamauchi normal form, with

$$a = -\frac{32(s-3)^4}{9(s+1)^3(s+3)^2}, \quad b = -\frac{8(s-3)^2(s-1)}{(s+1)(s+3)^2}.$$

Thus their signed parameter is  $u = s = ct$ , and in particular  $u^2 = 5t^2$ . Substitution in the standard formulas for  $c_4$  and  $\Delta$  yields

$$j = \frac{9(3t^2+1)(27t^2+1)^3}{t^2}. \quad (3.1)$$

The Fourier coefficients, diagonal normalization, and invariant calculation are replayed exactly in the paper's verification bundle.

Putting  $T = 81t^2$ , equation (3.1) becomes

$$j = J(T) = \frac{(T+27)(T+3)^3}{T}. \quad (3.2)$$

This is the standard degree-four  $j$ -map on  $X_0(3)$ . A convenient Tate normal form is

$$E_T : \quad y^2 + (T+27)xy + (T+27)^2y = x^3. \quad (3.3)$$

The point  $(0, 0)$  has order three, and the standard Weierstrass formulas give

$$\Delta(E_T) = T(T + 27)^8, \quad j(E_T) = J(T). \quad (3.4)$$

The equality of  $j$ -maps alone would identify only a generic elliptic isomorphism class. The next proposition supplies the geometric comparison needed for the principal-kernel packet.

**Proposition 3.1** (Actual-axis comparison). *Over the common marked smooth  $A_5/D_5$  base, the van Geemen–Yamauchi Prym factor maps isomorphically, as a polarized elliptic scheme, to the primitive dihedral norm axis inside the intermediate Jacobian. In particular its two-torsion local system is the  $\mathcal{E}[2]$  occurring in (1.1). Relative to (3.3), it may differ by the quadratic twist*

$$D(T) = (T + 27)(T - 729/5),$$

which does not alter its two-torsion.

*Proof.* Let  $\tilde{H} \rightarrow H$  be the étale double cover whose Prym is the intermediate Jacobian, and let  $\tilde{H}_0 \rightarrow H_0$  be its quotient by the cyclic subgroup of order five. The diagram in [9, Section 3.2] gives a pullback homomorphism

$$\phi : P_0 := \text{Prym}(\tilde{H}_0/H_0) \longrightarrow P := \text{Prym}(\tilde{H}/H) = J(X).$$

Write  $\Xi_0$  and  $\Xi$  for the principal Prym polarizations. Pullback and norm on the two Jacobians compose to multiplication by five. Moreover, the restriction of a Jacobian theta polarization to the Prym of an étale double cover is twice its principal Prym polarization. Restricting the Jacobian pullback identity to  $P_0$  therefore gives

$$2\phi^*\Xi = 10\Xi_0, \quad \text{hence} \quad \phi^*\Xi = 5\Xi_0. \quad (3.5)$$

The image of  $\phi$  is the connected fixed elliptic subvariety for the order-five subgroup: this is also the tangent-space calculation in [9, Proposition 3.2]. In the five-dimensional augmentation representation, that fixed line is fixed by its dihedral normalizer, so it is the primitive dihedral norm axis  $\mathcal{E}$ . Let  $i : \mathcal{E} \hookrightarrow \mathcal{J}$  be its inclusion and factor  $\phi = i\bar{\phi}$ . The diagonal entry of the six-axis polarization is five; equivalently, [8, Generic six-axis polarization and Relative six-axis source] gives

$$i^*\Xi = 5\lambda_{\mathcal{E}}.$$

Combining this with (3.5) gives

$$5\bar{\phi}^*\lambda_{\mathcal{E}} = 5\lambda_{P_0}.$$

The group of symmetric homomorphisms of elliptic schemes is torsion-free, so the factors five cancel. Taking degrees now forces  $\deg \bar{\phi} = 1$ . Thus  $\bar{\phi}$  is a polarized isomorphism on the open where the quotient diagram is smooth. Both sides are elliptic schemes over the regular marked base; the generic isomorphism and its inverse extend uniquely by the Néron mapping property. This also covers the smooth parameters at which an auxiliary quotient curve degenerates, as in the specialization argument of [9, Proposition 3.2].

It remains to identify the displayed model. If  $c_4^{\text{VGY}}, c_6^{\text{VGY}}$  and  $c_4^{\text{T}}, c_6^{\text{T}}$  denote the invariants of the two Weierstrass equations, the square class of their twist is represented by

$$q = \frac{c_6^{\text{VGY}} c_4^{\text{T}}}{c_4^{\text{VGY}} c_6^{\text{T}}}.$$

After  $T = 81u^2/5$ , direct factorization gives

$$\frac{q}{(T+27)(T-729/5)} = \left( \frac{200(u-3)^2}{2187(u+1)^2(u+3)^2(3u^2+5)} \right)^2.$$

This proves the asserted twist class. In characteristic different from two, a quadratic twist tensors the torsion representation by the scalar character  $\{\pm 1\}$ . It is therefore trivial on two-torsion and preserves every cyclic order-three subgroup, although it need not preserve a chosen generator.  $\square$

The proof also explains the role of earlier results. Roulleau exhibits the nonstandard  $A_5$ -pencil and its elliptic-power decomposition [6, Theorem 11(D) and Lemma 17]; Hartlieb places its intermediate Jacobians in a one-dimensional special subvariety [4, Lemma 5.5 and Proposition 5.7]. Neither statement by itself identifies the relative two-division packet used here.

## 4 The modular $S_3$ -diagram

Reduction modulo two gives

$$\rho_2 : \Gamma_0(3) \longrightarrow \mathrm{SL}_2(\mathbf{F}_2) \simeq S_3.$$

It is surjective: the elements

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$$

belong to  $\Gamma_0(3)$  and reduce to two distinct transvections, which generate  $\mathrm{SL}_2(\mathbf{F}_2)$ . Define

$$\Gamma_{\mathrm{sgn}}(6) = \ker(\mathrm{sgn} \circ \rho_2), \quad \Gamma(2, 3) = \Gamma(2) \cap \Gamma_0(3).$$

The first has index two and the second index six in  $\Gamma_0(3)$ . The resulting normalized Cartesian square is

$$\begin{array}{ccc} X(\Gamma(2, 3)) & \xrightarrow{2} & X_0(6) \\ 3 \downarrow & & \downarrow 3 \\ X_{\mathrm{sgn}}(6) & \xrightarrow{2} & X_0(3). \end{array} \tag{4.1}$$

A nonzero point of  $E[2]$  is the same as a cyclic subgroup of order two. Together with the given cyclic subgroup of order three, it determines a unique cyclic subgroup of order six. Conversely, an order-six cyclic subgroup has unique subgroups of orders two and three. Therefore the degree-three root resolvent is

$$X_0(6) \longrightarrow X_0(3).$$

The sign resolvent belongs to  $\Gamma_{\mathrm{sgn}}(6)$ , and the full ordered two-torsion cover belongs to  $\Gamma(2, 3)$ . This proves the group part of Theorem 1.2(3).

All three covers are rational. With  $y$  on the root cover,  $r$  on the sign cover, and  $v$  on the common splitting cover, one choice of equations is

$$T = -\frac{(4y+3)(y+3)^2}{(y+1)^2}, \quad (4.2)$$

$$x = -\frac{4y^4}{(y+1)^2}, \quad (4.3)$$

$$r^2 = T, \quad (4.4)$$

$$y = -\frac{v^2+3}{4}, \quad T = \frac{v^2(9-v^2)^2}{(1-v^2)^2}, \quad r = \frac{v(9-v^2)}{1-v^2}. \quad (4.5)$$

Substitution of (4.2) and (4.3) in the two-division cubic below gives zero, so  $y$  really selects a nonzero two-torsion point. The last line directly verifies the root and sign cover equations. It realizes the classical fact that adjoining a root of an  $S_3$ -cubic and adjoining the square root of its discriminant generate its splitting field.

**Corollary 4.1** (Golden orientation cyclicizes the rational packet). *After base change to the sign curve, the root cover  $X_0(6) \rightarrow X_0(3)$  becomes a connected Galois cover of degree three with group  $A_3 \simeq C_3$ . It is the full splitting cover of the two-division cubic. Hence, over the signed smooth cubic base, the rational three-sheet gluing packet has cyclic monodromy and any one rational sheet determines the other two.*

*Proof.* The sign subgroup has mod-two image  $A_3$ , while a point stabilizer in  $S_3$  has order two. Their intersection is trivial. Therefore the pullback of the point-stabilizer cover to the sign cover has regular  $A_3$ -action on its three sheets. Equivalently, the fibre product of (4.2) and (4.4) is the splitting field parametrized by (4.5).  $\square$

**Corollary 4.2** (Kummer normal form and local monodromy). *Let  $a^2 = -3$ . For the coordinates in (4.5),*

$$r^2 + 27 = \frac{(v^2+3)^3}{(v^2-1)^2}, \quad \frac{r-3a}{r+3a} = \left(\frac{v-a}{v+a}\right)^3. \quad (4.6)$$

*Thus the cyclic cubic cover of the signed line is totally ramified at  $v = \pm a$ , over  $r = \pm 3a$ , and nowhere else. These are precisely the two points  $t = \pm a/3$ , where  $t^2 = -1/3$  and the cubic has five  $A_2$ -points. The two local monodromies are inverse three-cycles; the mod-two monodromy around the two cusps and the two chordal points is trivial. The deck group is already rational over  $\mathbf{Q}(r)$ : it is generated by*

$$\sigma(v) = \frac{v-3}{v+1}, \quad \sigma^2(v) = -\frac{v+3}{v-1}, \quad \sigma^3(v) = v. \quad (4.7)$$

*Reversing the golden orientation sends  $(r, v)$  to  $(-r, -v)$  and conjugates  $\sigma$  to  $\sigma^{-1}$ .*

*Proof.* Both identities follow by substitution in (4.5). Equivalently,

$$\frac{dr}{dv} = \frac{(v^2+3)^2}{(v^2-1)^2},$$

so  $v = \pm a$  are the only critical points and each has ramification index three. Their branch values are  $r = \pm 3a$ . The product relation on the punctured signed line makes the two three-cycles inverse. At the cusps, the split and sign curves have equal widths 2 or 6, so the map is unramified; it is also unramified at the ordinary chordal points. This proves Theorem 1.2(6). Direct substitution gives  $r(\sigma(v)) = r(v)$ , and the fixed-point equation for  $\sigma$  is  $v^2 = -3$ . Finally  $-\sigma(-v) = \sigma^{-1}(v)$ , which proves the orientation assertion.  $\square$

For completeness, the two-division cubic of (3.3), obtained by eliminating  $y$  from the equation and  $2y + (T + 27)x + (T + 27)^2 = 0$ , is

$$4x^3 + (T + 27)^2 x^2 + 2(T + 27)^3 x + (T + 27)^4.$$

Its discriminant is

$$16T(T + 27)^8. \quad (4.8)$$

Thus its discriminant square class is exactly  $T$ . By Proposition 3.1 and Lemma 2.1, the exotic pair in the actual gluing packet is (4.4). Pulling back along  $T = 81t^2$  splits it; choosing one sheet gives  $r = 9t$ . This completes the proof of Theorem 1.2(1)–(3). Over the sign curve the mod-two image is  $A_3$ , by construction of  $\Gamma_{\text{sgn}}(6)$ . Passing to the smaller smooth cubic open cannot decrease that image: concretely, a small loop around either deleted lift of the order-three point  $T = -27$  acts as a three-cycle. Proposition 3.1 transports this action to the actual norm axis. This proves assertion (5).

## 5 Compactification and the cubic boundary

The modular orbifold attached to  $\Gamma_0(3)$  has two cusps of widths one and three and one elliptic point of order three. A primitive parabolic at either cusp reduces modulo two to a transposition, so it does not lie in  $\Gamma_{\text{sgn}}(6)$ , while its square does. Each cusp consequently has one ramified lift, of width two or six. An order-three elliptic generator reduces to a three-cycle and lies in the sign kernel; the elliptic point has two unramified lifts, both of order three.

The sign subgroup has index eight in  $\text{PSL}_2(\mathbf{Z})$ . The genus formula now reads

$$g = 1 + \frac{8}{12} - \frac{0}{4} - \frac{2}{3} - \frac{2}{2} = 0.$$

This proves Theorem 1.2(4), including the absence of any further coarse branching.

The same inertia calculation completes the passport of (4.1). On the root cover, a transposition has orbits of sizes one and two, giving cusp widths 1, 2, 3, 6. On the regular split cover it has three orbits of size two, giving widths 2, 2, 2, 6, 6, 6. The root and split subgroups contain no elliptic torsion. Their indices in  $\text{PSL}_2(\mathbf{Z})$  are twelve and twenty-four, so the genus formula gives genus zero in both cases. Thus every curve in the resolvent square is rational, now independently of the displayed parametrizations.

The eta Hauptmodul

$$h(\tau) = \left( \frac{\eta(\tau)}{\eta(3\tau)} \right)^{12}$$

has a simple pole at the width-one cusp and a simple zero at the width-three cusp. In the normalization (3.2),

$$h = \frac{729}{T}.$$

Because  $729 = 27^2$ , the quadratic fields  $\mathbf{Q}(h)(\sqrt{h})$  and  $\mathbf{Q}(T)(\sqrt{T})$  coincide. Analytically one may take  $\sqrt{h} = (\eta(\tau)/\eta(3\tau))^6$ . Its half-integral order in the base cusp parameter is the width doubling just computed.

Combining this square root with  $r = 9t$  gives an explicit modular uniformization of the signed cubic parameter:

$$\boxed{t(\tau) = 3 \left( \frac{\eta(3\tau)}{\eta(\tau)} \right)^6} \quad (5.1)$$

after the same deck convention. In the  $h$ -coordinate, the two noncuspidal cubic boundary values below are especially simple:

$$T = -27 \iff h = -27, \quad T = 729/5 \iff h = 5.$$

It remains to compare compactifications. A direct singularity calculation for the six-variable cubic equation gives

| cubic value  | $T$ -value   | degeneration        |
|--------------|--------------|---------------------|
| $t = \infty$ | $T = \infty$ | six singular points |
| $t = 0$      | $T = 0$      | ten singular points |
| $t^2 = -1/3$ | $T = -27$    | five $A_2$ -points  |
| $t^2 = 9/5$  | $T = 729/5$  | chordal cubic       |

**Proposition 5.1** (The chordal elliptic factor). *At  $T = 729/5$ , the actual transverse term of the pencil cuts a reduced degree-twelve divisor on the singular rational normal curve of the chordal cubic. The limit intermediate Jacobian is the Jacobian of the icosahedral hyperelliptic curve*

$$C_{\text{ico}} : w^2 = x(x^{10} + 11x^5 - 1).$$

Moreover

$$J(C_{\text{ico}}) \sim E_{\text{ico}}^5, \quad E_{\text{ico}} : y^2 = x(x^2 + 11x - 1),$$

and the elliptic factor has

$$j(E_{\text{ico}}) = \frac{2^{14} 31^3}{5^3} = J\left(\frac{729}{5}\right).$$

Consequently the modular value  $h = 5$  records the elliptic isogeny factor of the hyperelliptic chordal limit.

*Proof.* It is enough to treat one of the two marked points above  $T = 729/5$ , since the outer normalizer exchanges them. Let  $\zeta$  be a primitive fifth root of unity and put

$$p = [1 : \zeta : \zeta^2 : \zeta^3 : \zeta^4 : 0] \in \mathbf{P}(W).$$

For

$$t_0 = -\frac{3}{5} - \frac{6}{5}(\zeta^2 + \zeta^3), \quad t_0^2 = \frac{9}{5},$$

the Fourier scalar above is  $c = 1 + 2(\zeta^2 + \zeta^3)$ , so  $ct_0 = -3$ . The Fourier equation at this value is proportional to

$$F_{\text{ch}} = xu^2 + zv^2 - 2yuv - 4y^3 + 4xyz.$$

It has the catalecticant form

$$F_{\text{ch}} = 4 \det \begin{pmatrix} x & iv/2 & y \\ iv/2 & y & iu/2 \\ y & iu/2 & z \end{pmatrix}.$$

For completeness, write the five Hankel coordinates as  $(a, b, c_0, d, e)$ . The gradient of

$$ac_0e + 2bc_0d - ad^2 - b^2e - c_0^3$$

vanishes precisely when

$$c_0e = d^2, \quad c_0d = be, \quad bc_0 = ad, \quad ac_0 = b^2, \quad ae + 2bd = 3c_0^2.$$

If  $c_0 \neq 0$ , the first four equations give  $ae = bd$ , and the last then gives  $ae = bd = c_0^2$ . If  $c_0 = 0$ , they give  $b = d = 0$  and  $ae = 0$ . Thus the reduced singular locus is the rank-one Hankel locus

$$[a : b : c_0 : d : e] = [r^4 : r^3s : r^2s^2 : rs^3 : s^4],$$

a rational normal quartic, and the cubic is its secant variety. This proves directly that  $t_0$  is the asserted chordal value.

Direct differentiation gives  $p \in \text{Sing}(X_{t_0})$  and  $Q(p) = 0$ . Its projective  $A_5$ -orbit  $Z$  has twelve points, all lying on the reduced singular rational normal quartic  $C = \text{Sing}(X_{t_0})_{\text{red}}$ , and  $Q$  vanishes on all of them.

The exact linear-algebra calculation is short. Evaluation of the fifteen quadrics on  $Z$  has rank nine, so  $I_Z(2)$  has dimension six. Since a rational normal quartic also has  $\dim I_C(2) = 15 - 9 = 6$ , the inclusion  $I_C(2) \subset I_Z(2)$  is equality. The ideal of a rational normal curve is generated by its quadrics. In degree three their linear multiples span a space of dimension twenty-two, whereas adjoining  $Q$  raises the rank to twenty-three. Thus  $Q \notin I_C(3)$ .

Consequently  $Q|_C$  is a nonzero section of  $\mathcal{O}_C(3) \simeq \mathcal{O}_{\mathbf{P}^1}(12)$ . Its zero divisor contains the twelve distinct points of  $Z$ , so it is exactly  $Z$  and is reduced. The first-order deformation of  $P + t_0Q$  in this pencil is  $Q$ ; hence the nonvanishing criterion of Allcock–Carlson–Toledo shows that this direction is transverse to the chordal locus [1, Lemma 2.5]. The general chordal-limit description therefore applies [3, Remark 1.8]. The icosahedral action on  $C \simeq \mathbf{P}^1$  has a unique orbit of length twelve. In a standard coordinate it is the zero divisor of

$$su(s^{10} + 11s^5u^5 - u^{10}),$$

which gives the displayed equation for  $C_{\text{ico}}$ .

Paulhus proves the displayed isogeny decomposition and elliptic factor [7, Theorem 2]. For the printed Weierstrass equation,  $c_4 = 1984 = 2^6 \cdot 31$  and  $\Delta = 2000 = 2^4 \cdot 5^3$ , hence  $j(E_{\text{ico}}) = 2^{14}31^3/5^3$ . Direct substitution in  $J(T) = (T+27)(T+3)^3/T$  gives the same value at  $T = 729/5$ .  $\square$

**Corollary 5.2** (Boundary reconstruction). *The embedded branch orbit  $Z \subset \mathbf{P}(W)$  determines the pair  $(C, [Q|_C])$ , hence the chordal central fibre and its projective first-order normal direction, which is the unique  $A_5$ -fixed normal line. Thus all  $A_5$ -invariant transverse directions have the same hyperelliptic limit.*

*Proof.* Quadratic generation and  $I_Z(2) = I_C(2)$  recover  $C$  as a scheme; its secant variety is the chordal cubic. The divisor  $Z$  uniquely determines  $[Q|_C] \in \mathbf{P}H^0(C, \mathcal{O}_C(3))$ . The equivariant normal-space calculation in [1, Lemma 2.5] identifies this section with the projective normal line  $[Q]$  modulo  $I_C(3)$ . Conversely, an  $A_5$ -fixed projective normal line is represented by a semi-invariant section. Since  $A_5$  is perfect, it has no nontrivial complex character, so the section is invariant. Its zero divisor is therefore an invariant effective divisor of degree twelve on  $C \simeq \mathbf{P}^1$ . The nontrivial icosahedral orbit sizes are 12, 20, 30, 60; hence that divisor is the unique reduced orbit  $Z$  of length twelve.  $\square$

Only  $T = 0$  and  $T = \infty$  are cusps of  $X_0(3)$ . The marked smooth cubic base is therefore

$$\mathbf{P}_t^1 \setminus \{0, \infty, t^2 = -1/3, t^2 = 9/5\},$$

and its unmarked image is

$$X_0(3) \setminus \{0, \infty, -27, 729/5\}.$$

On these opens,  $t \mapsto T = 81t^2$  is finite *étale*. Its compactification ramifies only over the two modular cusps. This proves the boundary assertion of Theorem 1.2 and prevents a misleading identification of every cubic degeneration with a cusp.

## 6 Conclusion and scope

The object behind the five principal kernels is the  $S_3$ -torsor of ordered nonzero two-torsion on the actual relative norm axis. Its two proper transitive quotient sets are

$$S_3/C_2 \quad (3 \text{ sheets}), \quad S_3/A_3 \quad (2 \text{ sheets}).$$

Thus the golden pair contains no independent monodromy datum: it is the discriminant orientation of the rational root triple. The familiar decomposition  $\mathbf{P}^1(\mathbf{F}_4) = \mathbf{P}^1(\mathbf{F}_2) \sqcup \{\omega, \omega^2\}$  is the fibrewise form of this global resolvent theorem.

The modular identification here is deliberately limited. It neither turns the five principal kernels into ordinary elliptic isogenies nor promotes the whole compactified cubic pencil to a modular stack. In particular  $T = -27,729/5$  are cubic boundary points, not modular cusps; at the latter, Proposition 5.1 identifies the hyperelliptic limit. The related level-three period map for the Winger pencil [5, Theorem 1.1 and Proposition 5.1] concerns a distinct family and is not used here. The result instead combines the explicit cubic period coordinate, the polarized actual-axis comparison, and the principal-kernel local system.

**Reproducibility.** The directory `verification/` contains deterministic exact certificates for the resolvent identities, the chordal orbit and ideal ranks, and the subgroup enumeration, together with expected outputs and a SHA-256 manifest. The `make check` target replays them, verifies the manifest, lints the source, builds the PDF, and rejects TeX warnings.

## References

- [1] D. Allcock, J. A. Carlson, and D. Toledo, *The moduli space of cubic threefolds as a ball quotient*, Mem. Amer. Math. Soc. 209 (2011), no. 985; doi:10.1090/S0065-9266-10-00591-0.
- [2] M. Rebolledo and C. Wuthrich, *A moduli interpretation for the non-split Cartan modular curve*, Glasg. Math. J. 60 (2018), no. 2, 411–434.
- [3] S. Casalaina-Martin, S. Grushevsky, K. Hulek, and R. Laza, *Complete moduli of cubic threefolds and their intermediate Jacobians*, Proc. Lond. Math. Soc. 122 (2021), 259–316.
- [4] M. Hartlieb, *Special subvarieties in the locus of intermediate Jacobians of cubic threefolds*, Math. Z. 310 (2025), article 52.
- [5] E. Looijenga and Y. Zi, *Monodromy and period map of the Winger pencil*, Nagoya Math. J. 251 (2023), 561–602.
- [6] X. Roulleau, *The Fano surface of the Klein cubic threefold*, J. Math. Kyoto Univ. 49 (2009), no. 1, 113–129.
- [7] J. Paulhus, *Elliptic factors in Jacobians of hyperelliptic curves with certain automorphism groups*, Open Book Ser. 1 (2013), 487–505; doi:10.2140/obs.2013.1.487.
- [8] T. Rudd, *Irrationality of cubic threefolds after one stabilization*, preprint, 2026.
- [9] B. van Geemen and T. Yamauchi, *On intermediate Jacobians of cubic threefolds admitting an automorphism of order five*, Pure Appl. Math. Q. 12 (2016), no. 1, 141–164.